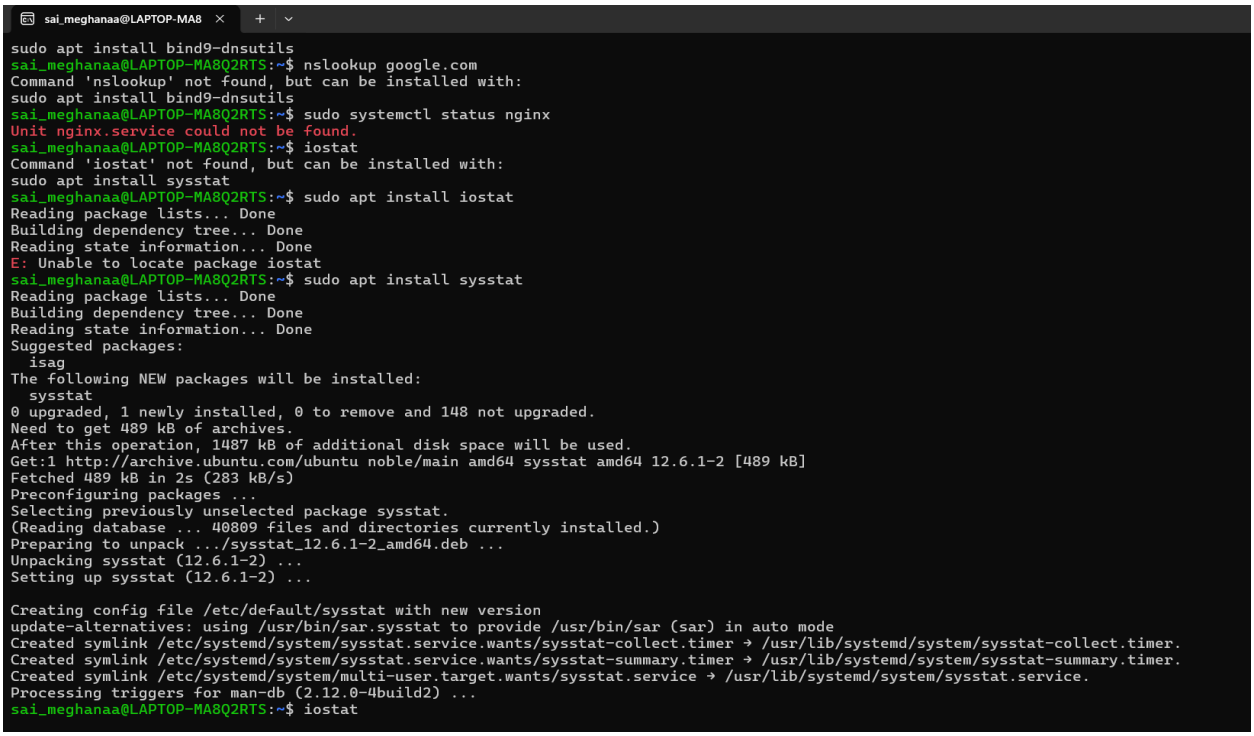
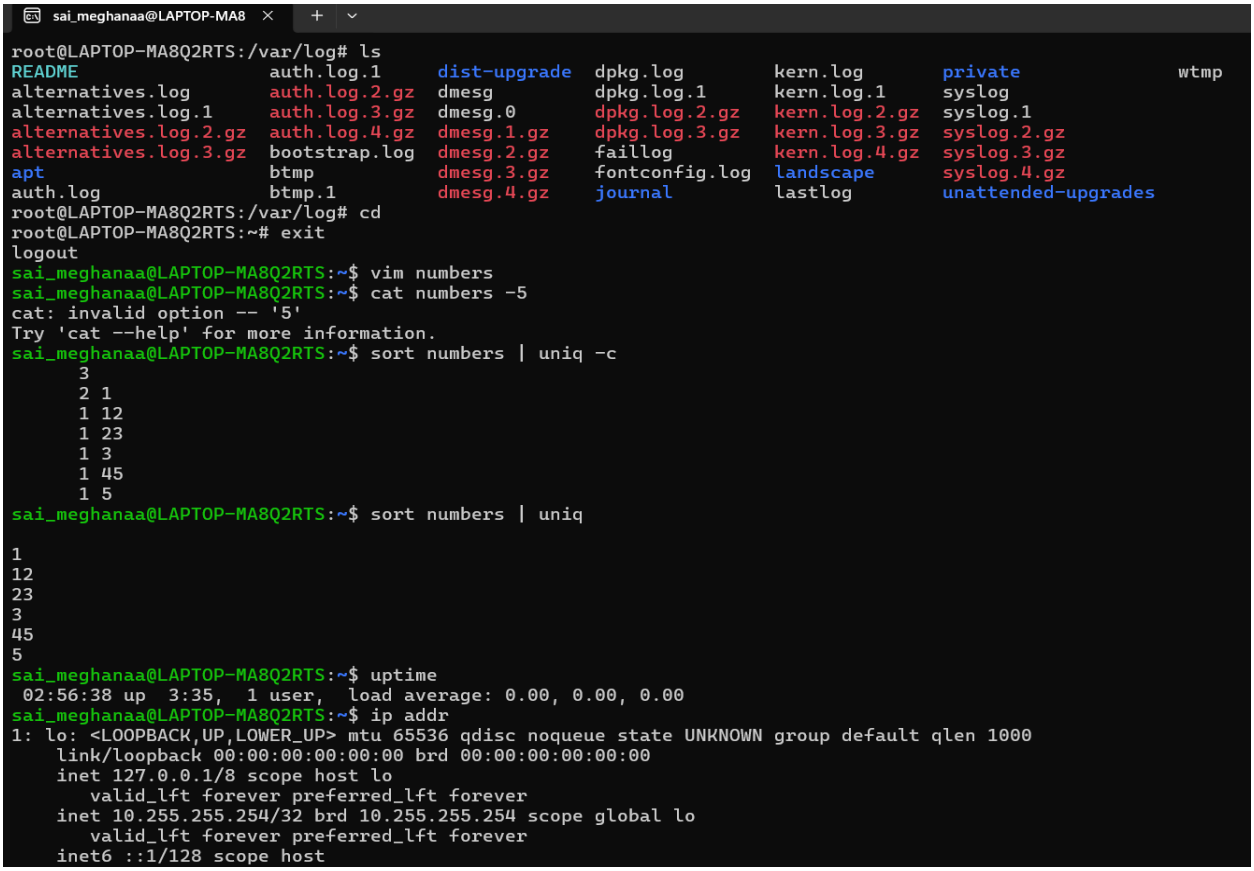
10 Basic Linux Commands:



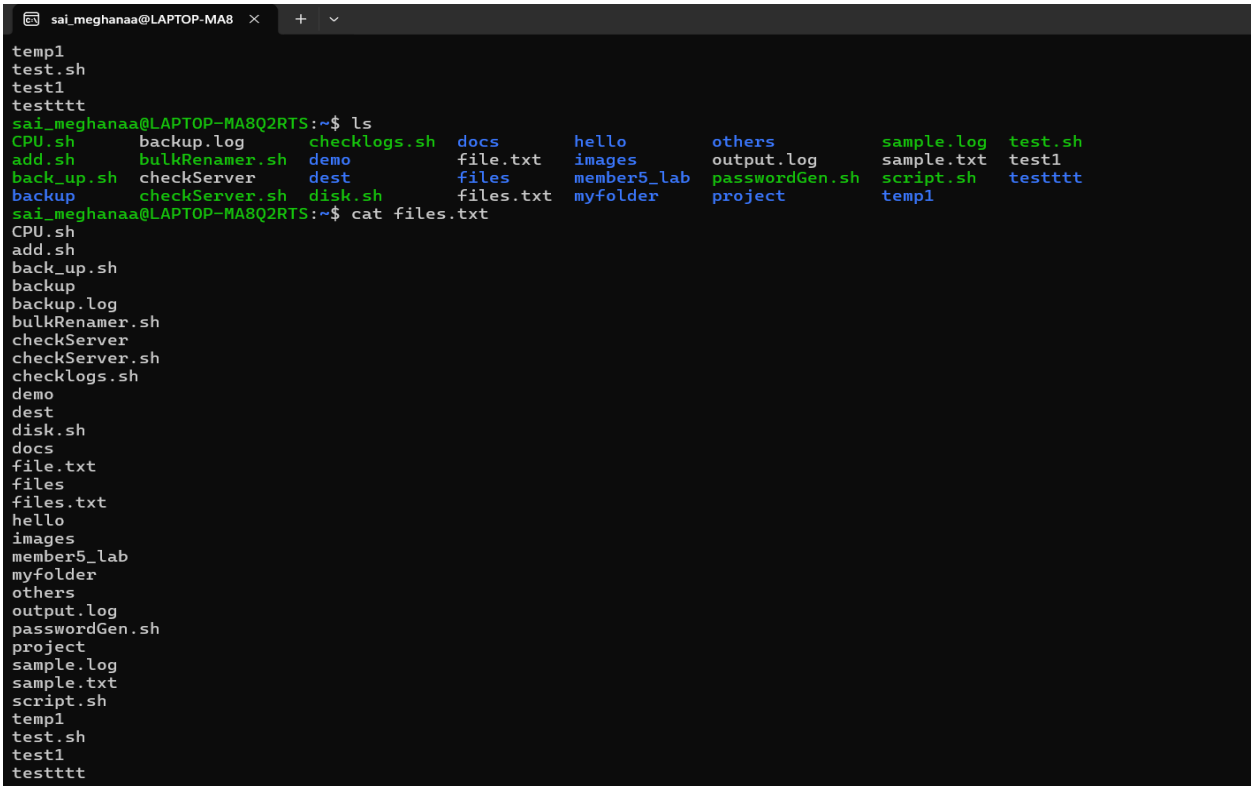
```
sai_meghanaa@LAPTOP-MA8
sudo apt install bind9-dnsutils
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ nslookup google.com
Command 'nslookup' not found, but can be installed with:
sudo apt install bind9-dnsutils
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sudo systemctl status nginx
Unit nginx.service could not be found.
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ iostat
Command 'iostat' not found, but can be installed with:
sudo apt install sysstat
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sudo apt install iostat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package iostat
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sudo apt install sysstat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  isag
The following NEW packages will be installed:
  sysstat
0 upgraded, 1 newly installed, 0 to remove and 148 not upgraded.
Need to get 489 kB of archives.
After this operation, 1487 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 sysstat amd64 12.6.1-2 [489 kB]
Fetched 489 kB in 2s (283 kB/s)
Preconfiguring packages ...
Selecting previously unselected package sysstat.
(Reading database ... 40809 files and directories currently installed.)
Preparing to unpack .../sysstat_12.6.1-2_amd64.deb ...
Unpacking sysstat (12.6.1-2) ...
Setting up sysstat (12.6.1-2) ...

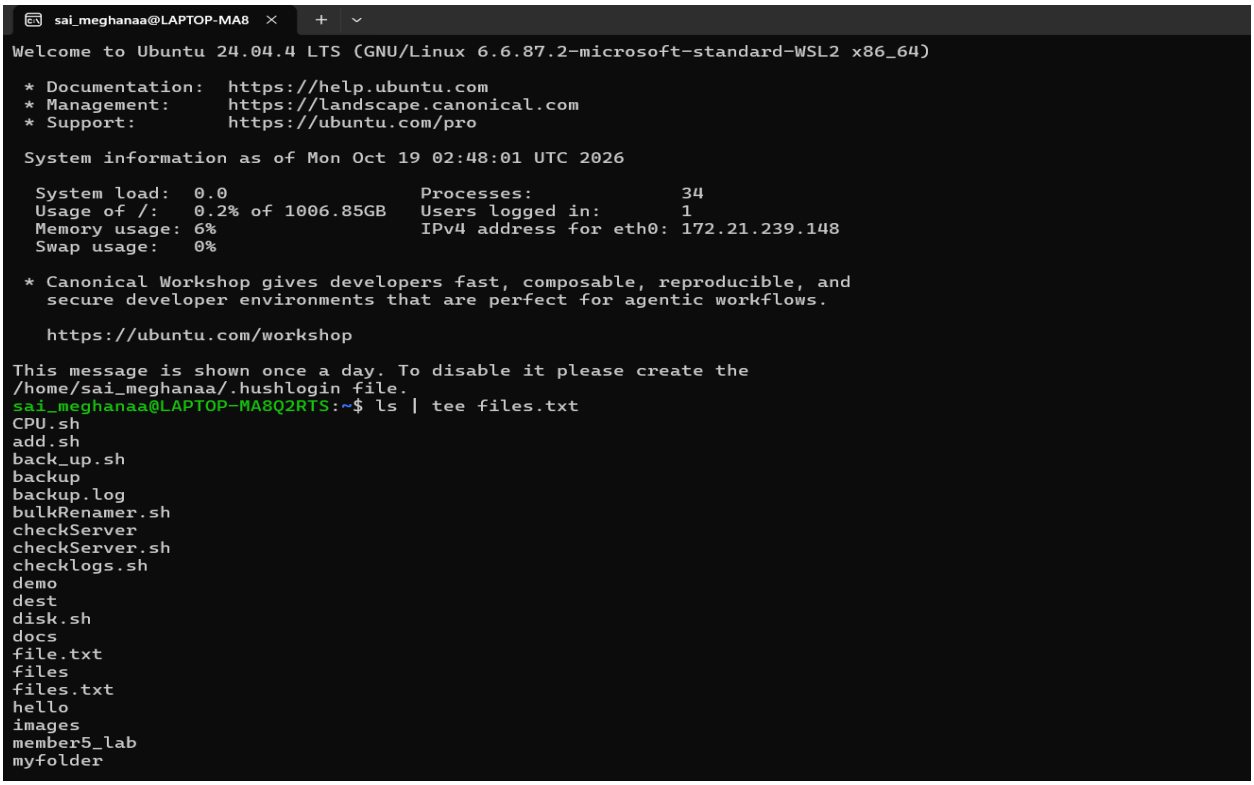
Creating config file /etc/default/sysstat with new version
update-alternatives: using /usr/bin/sar.sysstat to provide /usr/bin/sar (sar) in auto mode
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-collect.timer → /usr/lib/systemd/system/sysstat-collect.timer.
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-summary.timer → /usr/lib/systemd/system/sysstat-summary.timer.
Created symlink /etc/systemd/system/multi-user.target.wants/sysstat.service → /usr/lib/systemd/system/sysstat.service.
Processing triggers for man-db (2.12.0-4build2) ...
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ iostat
```

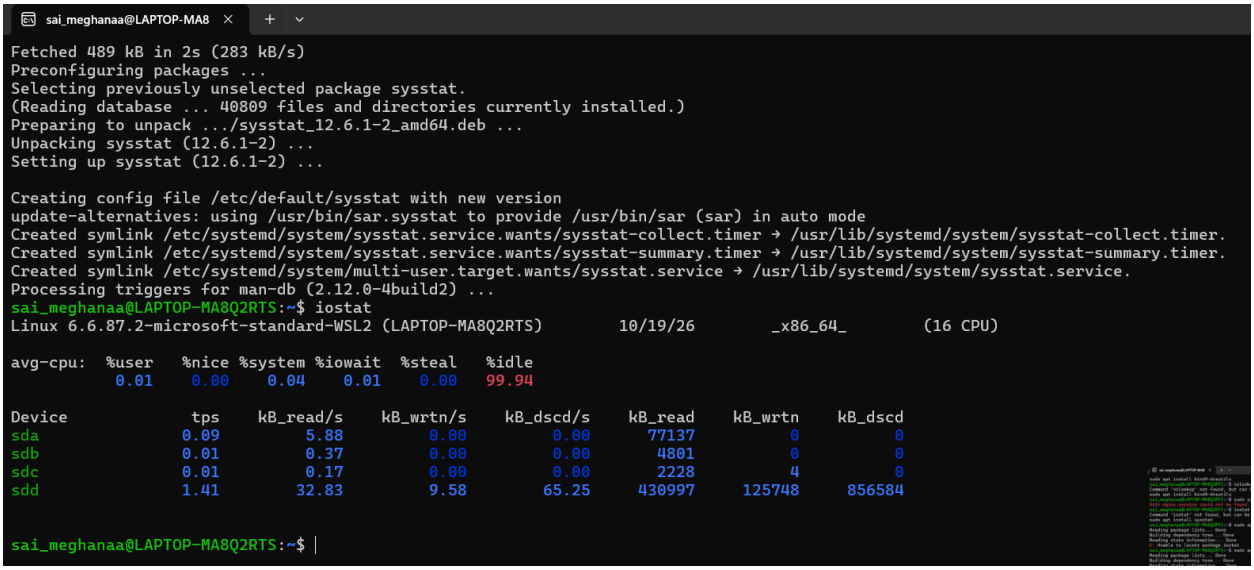


```
sai_meghanaa@LAPTOP-MA8
root@LAPTOP-MA8Q2RTS:/var/log# ls
README                  auth.log.1      dist-upgrade  dpkg.log        kern.log        private              wtmp
alternatives.log        auth.log.2.gz   dmesg         dpkg.log.1      kern.log.1      syslog
alternatives.log.1      auth.log.3.gz   dmesg.0       dpkg.log.2.gz   kern.log.2.gz   syslog.1
alternatives.log.2.gz   auth.log.4.gz   dmesg.1.gz    dpkg.log.3.gz   kern.log.3.gz   syslog.2.gz
alternatives.log.3.gz   bootstrap.log   dmesg.2.gz    faillog         kern.log.4.gz   syslog.3.gz
apt                     btmp            dmesg.3.gz    fontconfig.log  landscape       syslog.4.gz
auth.log                btmp.1          dmesg.4.gz    journal         lastlog         unattended-upgrades
root@LAPTOP-MA8Q2RTS:/var/log# cd
root@LAPTOP-MA8Q2RTS:~# exit
logout
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ vim numbers
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ cat numbers -5
cat: invalid option -- '5'
Try 'cat --help' for more information.
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sort numbers | uniq -c
      3
      2 1
      1 12
      1 23
      1 3
      1 45
      1 5
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ sort numbers | uniq

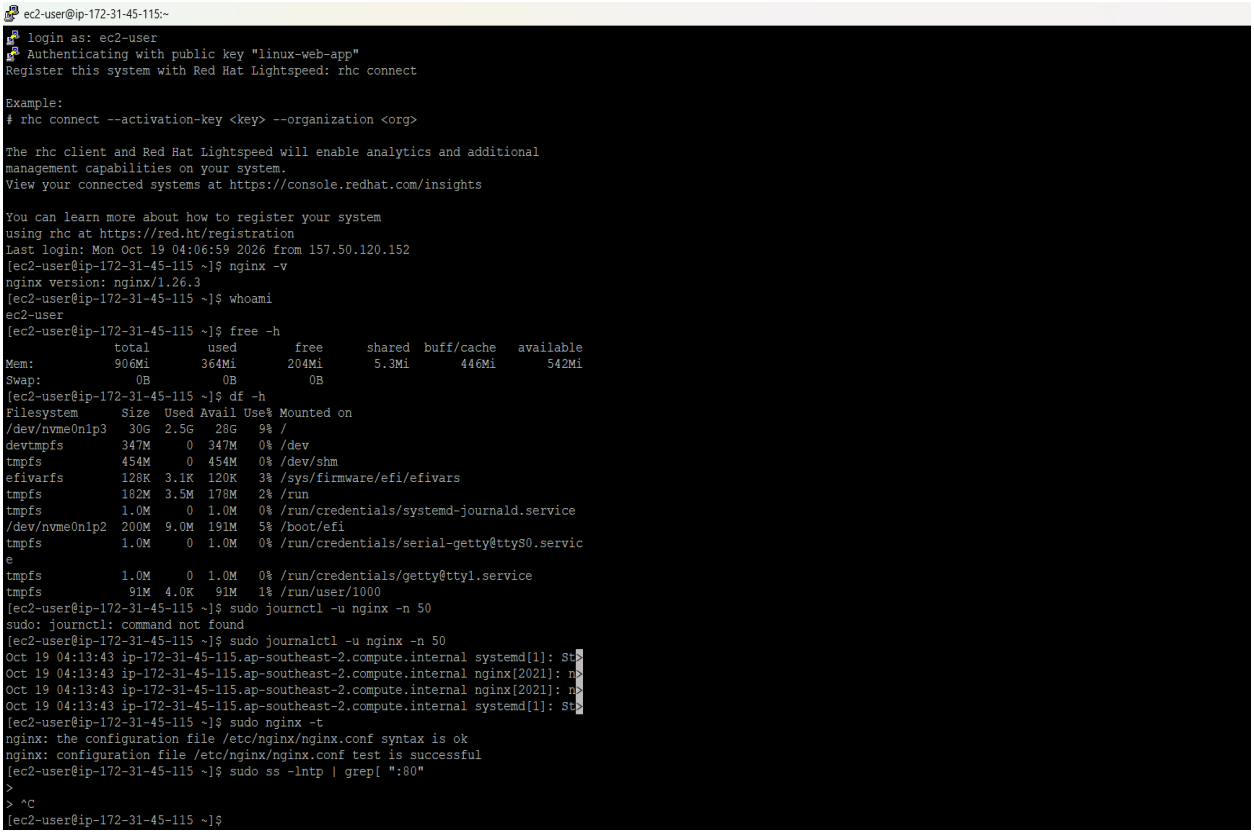
1
12
23
3
45
5
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ uptime
 02:56:38 up  3:35,  1 user,  load average: 0.00, 0.00, 0.00
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
```

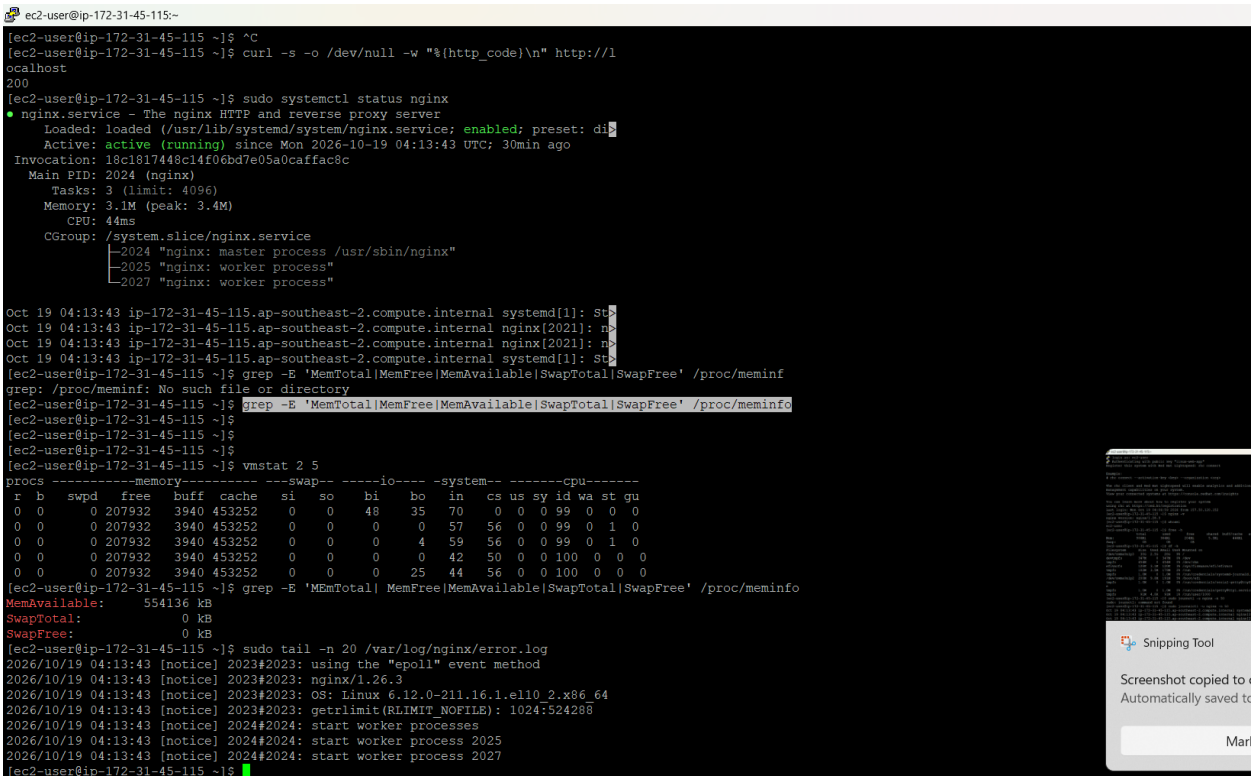

sai_meghanaa@LAPTOP-MA8
temp1
test.sh
test1
testttt
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ls
CPU.sh backup.log checklogs.sh docs hello others sample.log test.sh
add.sh bulkRenamer.sh demo file.txt images output.log sample.txt test1
back_up.sh checkServer dest files member5_lab passwordGen.sh script.sh testttt
backup checkServer.sh disk.sh files.txt myfolder project temp1
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ cat files.txt
CPU.sh
add.sh
back_up.sh
backup
backup.log
bulkRenamer.sh
checkServer
checkServer.sh
checklogs.sh
demo
dest
disk.sh
docs
file.txt
files
files.txt
hello
images
member5_lab
myfolder
others
output.log
passwordGen.sh
project
sample.log
sample.txt
script.sh
temp1
test.sh
test1
testttt


sai_meghanaa@LAPTOP-MA8
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Mon Oct 19 02:48:01 UTC 2026
System load: 0.0 Processes: 34
Usage of /: 0.2% of 1006.85GB Users logged in: 1
Memory usage: 6% IPv4 address for eth0: 172.21.239.148
Swap usage: 0%
* Canonical Workshop gives developers fast, composable, reproducible, and
secure developer environments that are perfect for agentic workflows.
https://ubuntu.com/workshop
This message is shown once a day. To disable it please create the
/home/sai_meghanaa/.hushlogin file.
sai_meghanaa@LAPTOP-MA8Q2RTS:~$ ls | tee files.txt
CPU.sh
add.sh
back_up.sh
backup
backup.log
bulkRenamer.sh
checkServer
checkServer.sh
checklogs.sh
demo
dest
disk.sh
docs
file.txt
files
files.txt
hello
images
member5_lab
myfolder

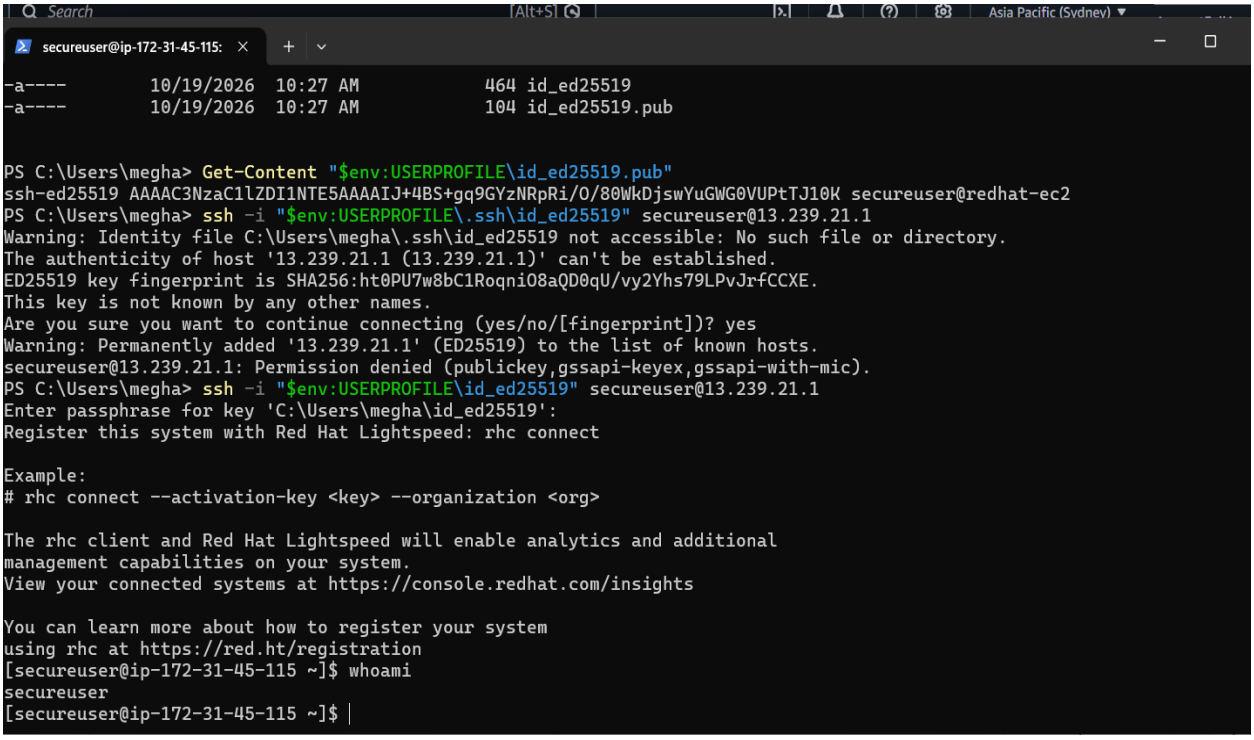


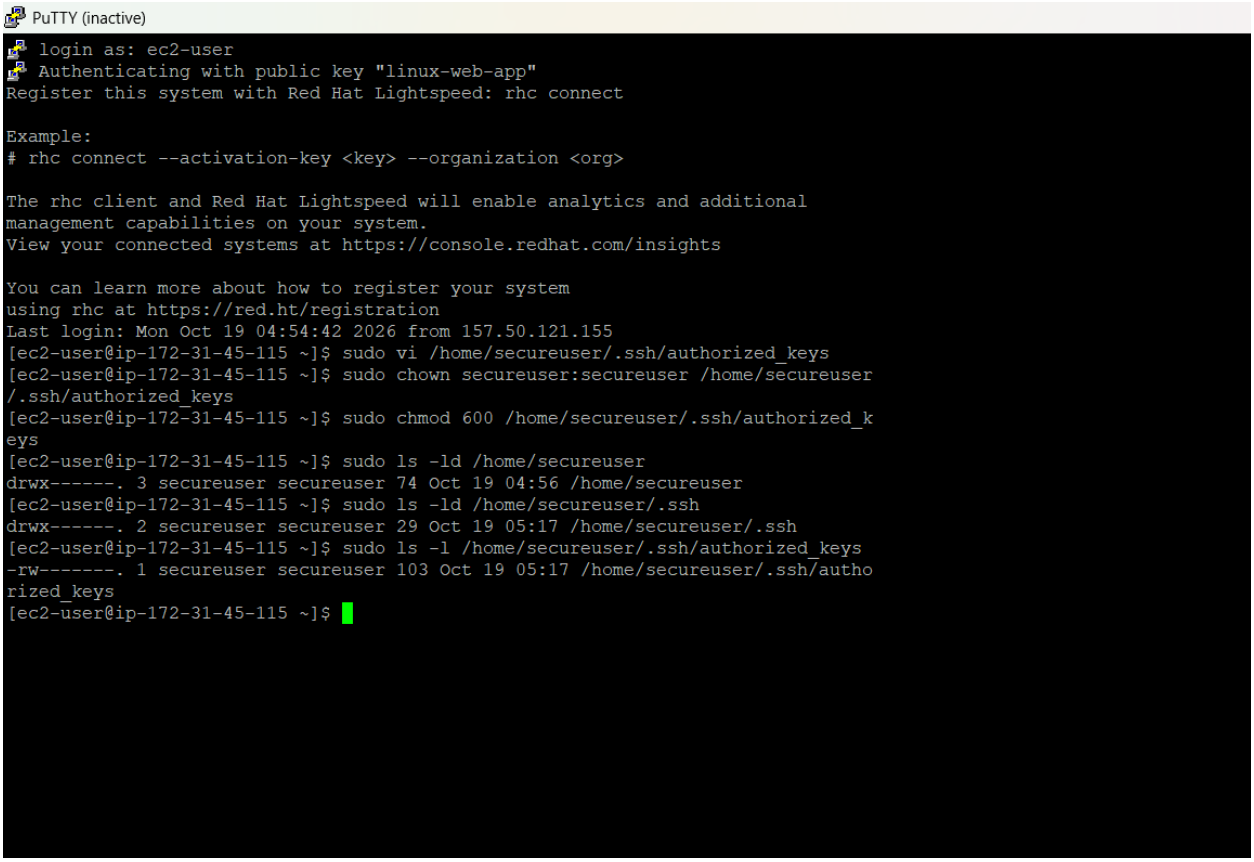
Commands to Check the memory and Configuration of nginx and then Complete the Excel of Top 10 check list:

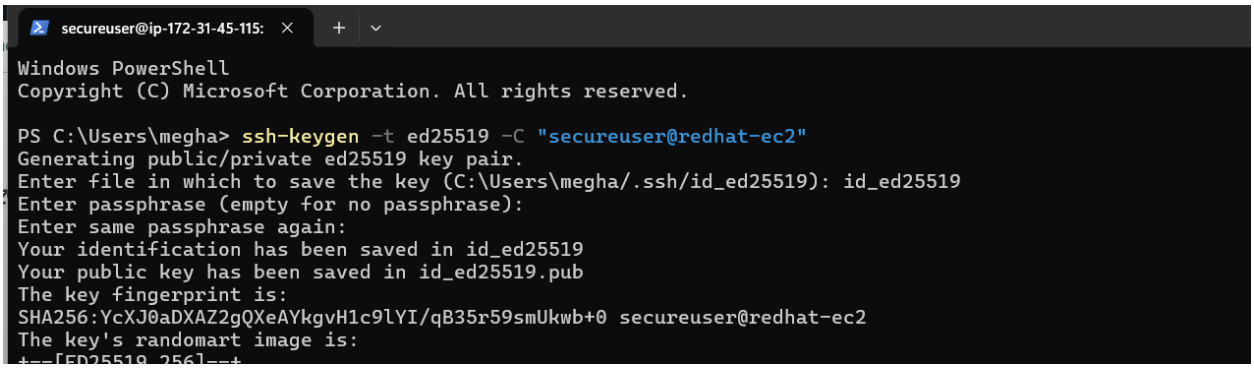




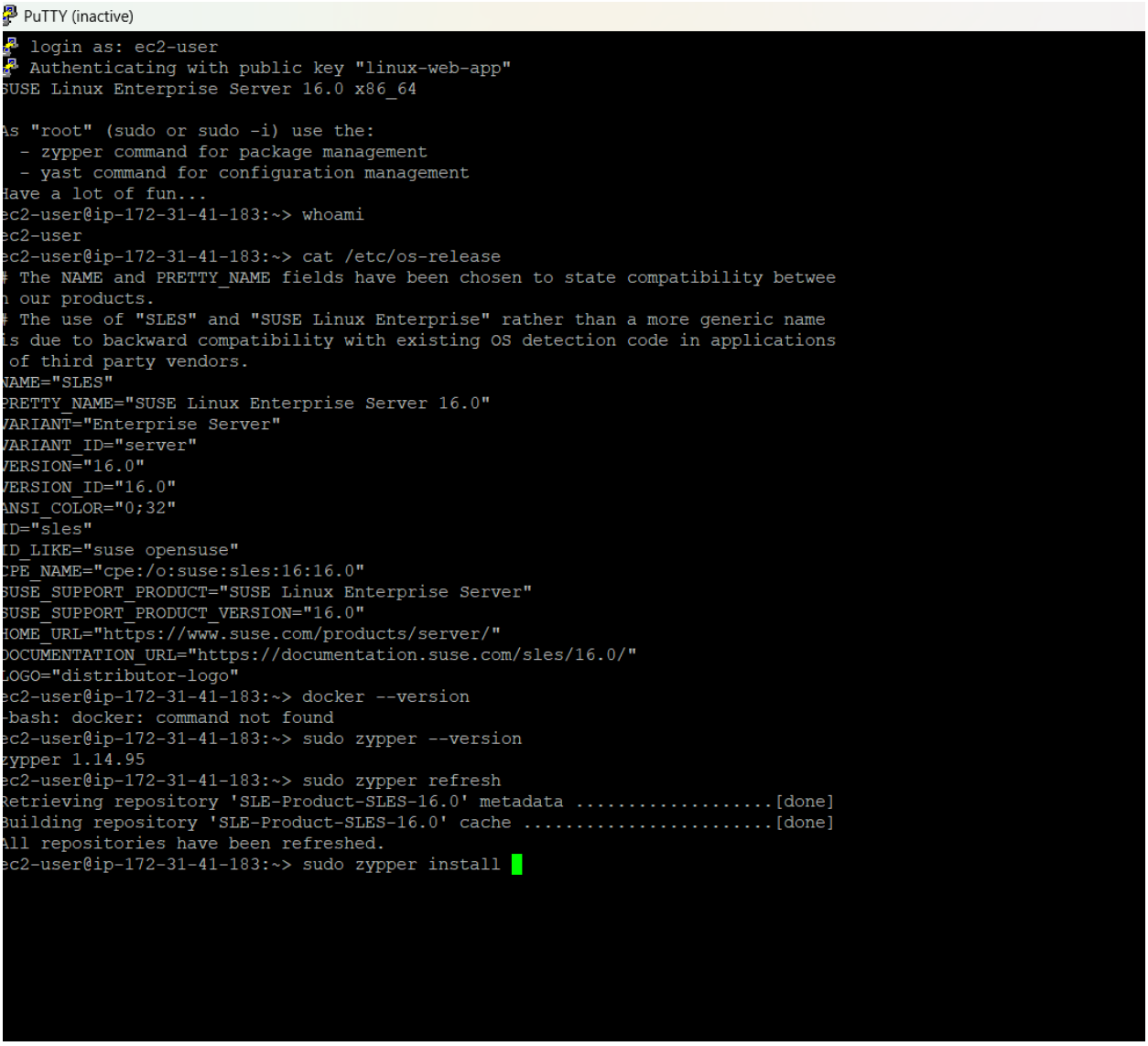
Create a Secured Connection to user in RedHat:

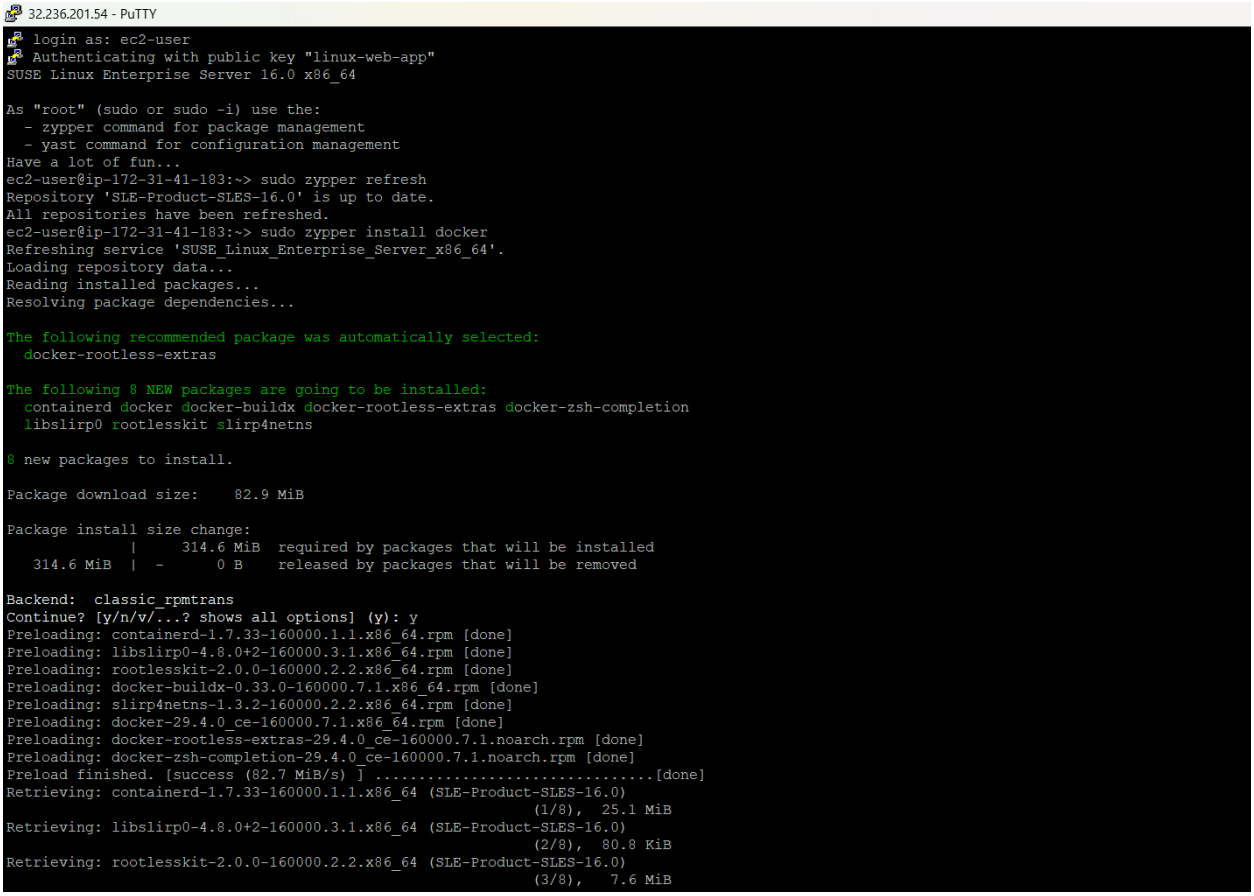



PuTTY (inactive)
login as: ec2-user
Authenticating with public key "linux-web-app"
Register this system with Red Hat Lightspeed: rhc connect
Example:
rhc connect --activation-key <key> --organization <org>
The rhc client and Red Hat Lightspeed will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights
You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Oct 19 04:54:42 2026 from 157.50.121.155
[ec2-user@ip-172-31-45-115 ~]$ sudo vi /home/secureuser/.ssh/authorized_keys
[ec2-user@ip-172-31-45-115 ~]$ sudo chown secureuser:secureuser /home/secureuser
/.ssh/authorized_keys
[ec2-user@ip-172-31-45-115 ~]$ sudo chmod 600 /home/secureuser/.ssh/authorized_k
eys
[ec2-user@ip-172-31-45-115 ~]$ sudo ls -ld /home/secureuser
drwx------. 3 secureuser secureuser 74 Oct 19 04:56 /home/secureuser
[ec2-user@ip-172-31-45-115 ~]$ sudo ls -ld /home/secureuser/.ssh
drwx------. 2 secureuser secureuser 29 Oct 19 05:17 /home/secureuser/.ssh
[ec2-user@ip-172-31-45-115 ~]$ sudo ls -l /home/secureuser/.ssh/authorized_keys
-rw-------. 1 secureuser secureuser 103 Oct 19 05:17 /home/secureuser/.ssh/autho
rized_keys
[ec2-user@ip-172-31-45-115 ~]$

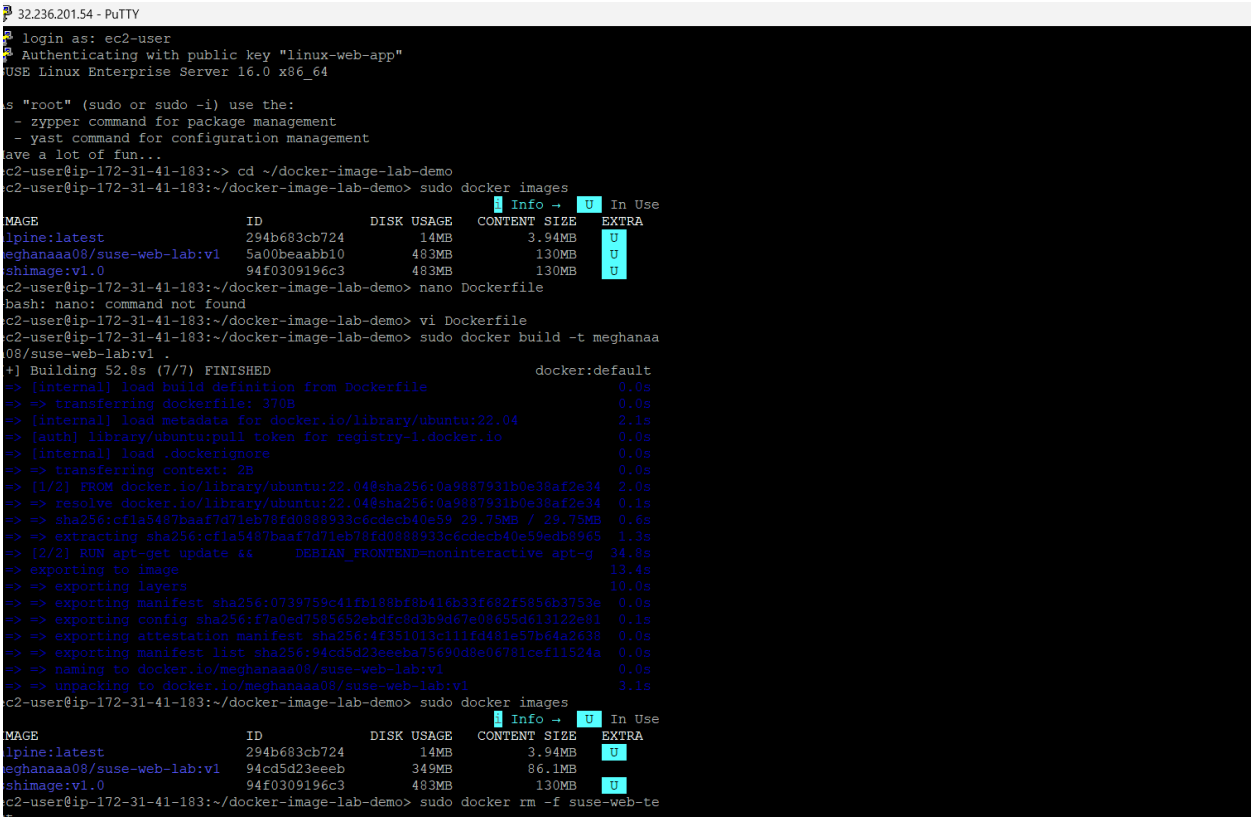

secureuser@ip-172-31-45-115:
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\megha> ssh-keygen -t ed25519 -C "secureuser@redhat-ec2"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\megha/.ssh/id_ed25519): id_ed25519
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in id_ed25519
Your public key has been saved in id_ed25519.pub
The key fingerprint is:
SHA256:YcXJ0aDXAZ2gQXeAYkgvH1c9lYI/qB35r59smUkwb+0 secureuser@redhat-ec2
The key's randomart image is:

Creating the Volumes and Check the Sync with the Local Volume and Local Directory:



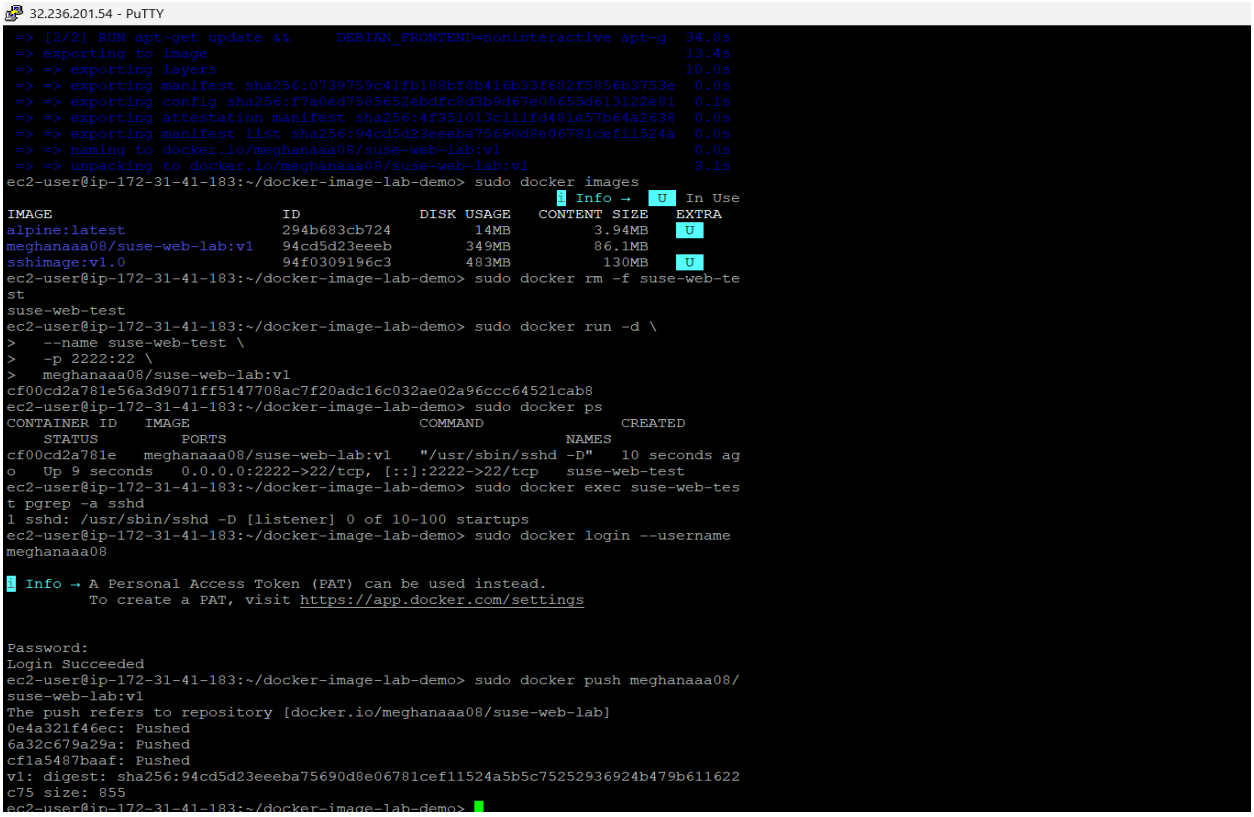

32.236.201.54 - PuTTY
login as: ec2-user
Authenticating with public key "linux-web-app"
SUSE Linux Enterprise Server 16.0 x86_64
As "root" (sudo or sudo -i) use the:
- zypper command for package management
- yast command for configuration management
Have a lot of fun...
ec2-user@ip-172-31-41-183:~> sudo zypper refresh
Repository 'SLE-Product-SLES-16.0' is up to date.
All repositories have been refreshed.
ec2-user@ip-172-31-41-183:~> sudo zypper install docker
Refreshing service 'SUSE_Linux_Enterprise_Server_x86_64'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...
The following recommended package was automatically selected:
docker-rootless-extras
The following 8 NEW packages are going to be installed:
containerd docker docker-buildx docker-rootless-extras docker-zsh-completion
libslirp0 rootlesskit slirp4netns
8 new packages to install.
Package download size: 82.9 MiB
Package install size change:
| 314.6 MiB required by packages that will be installed
314.6 MiB | - 0 B released by packages that will be removed
Backend: classic_rpmtrans
Continue? [y/n/v/...? shows all options] (y): y
Preloading: containerd-1.7.33-160000.1.1.x86_64.rpm [done]
Preloading: libslirp0-4.8.0+2-160000.3.1.x86_64.rpm [done]
Preloading: rootlesskit-2.0.0-160000.2.2.x86_64.rpm [done]
Preloading: docker-buildx-0.33.0-160000.7.1.x86_64.rpm [done]
Preloading: slirp4netns-1.3.2-160000.2.2.x86_64.rpm [done]
Preloading: docker-29.4.0_ce-160000.7.1.x86_64.rpm [done]
Preloading: docker-rootless-extras-29.4.0_ce-160000.7.1.noarch.rpm [done]
Preloading: docker-zsh-completion-29.4.0_ce-160000.7.1.noarch.rpm [done]
Preload finished. [success (82.7 MiB/s)] .....[done]
Retrieving: containerd-1.7.33-160000.1.1.x86_64 (SLE-Product-SLES-16.0)
(1/8), 25.1 MiB
Retrieving: libslirp0-4.8.0+2-160000.3.1.x86_64 (SLE-Product-SLES-16.0)
(2/8), 80.8 KiB
Retrieving: rootlesskit-2.0.0-160000.2.2.x86_64 (SLE-Product-SLES-16.0)
(3/8), 7.6 MiB

Creating the Image and Push to the DOcker Hub:



```
32.236.201.54 - PuTTY
login as: ec2-user
Authenticating with public key "linux-web-app"
USE Linux Enterprise Server 16.0 x86_64

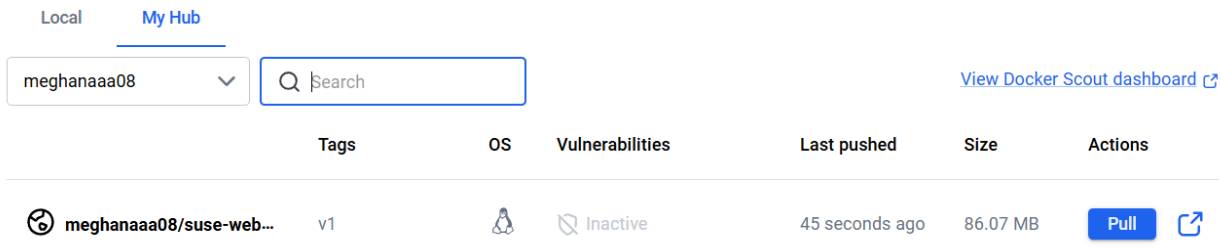
s "root" (sudo or sudo -i) use the:
  - zypper command for package management
  - yast command for configuration management
ave a lot of fun...
c2-user@ip-172-31-41-183:~> cd ~/docker-image-lab-demo
c2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker images
                                                    i Info →   U  In Use
MAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
lpine:latest                  294b683cb724         14MB         3.94MB    U
eghanaaa08/suse-web-lab:v1    5a00beaabb10        483MB          130MB    U
sshimage:v1.0                 94f0309196c3        483MB          130MB    U
c2-user@ip-172-31-41-183:~/docker-image-lab-demo> nano Dockerfile
bash: nano: command not found
c2-user@ip-172-31-41-183:~/docker-image-lab-demo> vi Dockerfile
c2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker build -t meghanaa
08/suse-web-lab:v1 .
+] Building 52.8s (7/7) FINISHED                                 docker:default
 => [internal] load build definition from Dockerfile                      0.0s
 => => transferring dockerfile: 370B                                      0.0s
 => [internal] load metadata for docker.io/library/ubuntu:22.04           2.1s
 => [auth] library/ubuntu:pull token for registry-1.docker.io             0.0s
 => [internal] load .dockerignore                                         0.0s
 => => transferring context: 2B                                           0.0s
 => [1/2] FROM docker.io/library/ubuntu:22.04@sha256:0a9887931b0e38af2e34  2.0s
 => => resolve docker.io/library/ubuntu:22.04@sha256:0a9887931b0e38af2e34  0.1s
 => => sha256:cf1a5487baaf7d71eb78fd0888933c6cdecb40e59 29.75MB / 29.75MB  0.6s
 => => extracting sha256:cf1a5487baaf7d71eb78fd0888933c6cdecb40e59edb8965  1.3s
 => [2/2] RUN apt-get update &&     DEBIAN_FRONTEND=noninteractive apt-g  34.8s
 => exporting to image                                                    13.4s
 => => exporting layers                                                   10.0s
 => => exporting manifest sha256:0739759c41fb188bf8b416b33f682f5856b3753e  0.0s
 => => exporting config sha256:f7a0ed7585652ebdfc8d3b9d67e08655d613122e81  0.1s
 => => exporting attestation manifest sha256:4f351013c111fd481e57b64a2638  0.0s
 => => exporting manifest list sha256:94cd5d23eeeba75690d8e06781cef11524a  0.0s
 => => naming to docker.io/meghanaaa08/suse-web-lab:v1                    0.0s
 => => unpacking to docker.io/meghanaaa08/suse-web-lab:v1                 3.1s
c2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker images
                                                    i Info →   U  In Use
MAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
lpine:latest                  294b683cb724         14MB         3.94MB    U
eghanaaa08/suse-web-lab:v1    94cd5d23eeeb        349MB         86.1MB
sshimage:v1.0                 94f0309196c3        483MB          130MB    U
c2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker rm -f suse-web-te
```



```
32.236.201.54 - PuTTY
 => [2/2] RUN apt-get update &&     DEBIAN_FRONTEND=noninteractive apt-g  34.8s
 => exporting to image                                                    13.4s
 => => exporting layers                                                   10.0s
 => => exporting manifest sha256:0739759c41fb188bf8b416b33f682f5856b3753e  0.0s
 => => exporting config sha256:f7a0ed7585652ebdfc8d3b9d67e08655d613122e81  0.1s
 => => exporting attestation manifest sha256:4f351013c111fd481e57b64a2638  0.0s
 => => exporting manifest list sha256:94cd5d23eeeba75690d8e06781cef11524a  0.0s
 => => naming to docker.io/meghanaaa08/suse-web-lab:v1                    0.0s
 => => unpacking to docker.io/meghanaaa08/suse-web-lab:v1                 3.1s
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker images
                                                    i Info →   U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                  294b683cb724         14MB         3.94MB    U
meghanaaa08/suse-web-lab:v1    94cd5d23eeeb        349MB         86.1MB
sshimage:v1.0                  94f0309196c3        483MB          130MB    U
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker rm -f suse-web-te
st
suse-web-test
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker run -d \
>   --name suse-web-test \
>   -p 2222:22 \
>   meghanaaa08/suse-web-lab:v1
cf00cd2a781e56a3d9071ff5147708ac7f20adc16c032ae02a96ccc64521cab8
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker ps
CONTAINER ID   IMAGE                          COMMAND              CREATED
    STATUS         PORTS                                  NAMES
cf00cd2a781e   meghanaaa08/suse-web-lab:v1    "/usr/sbin/sshd -D"   10 seconds ag
o   Up 9 seconds   0.0.0.0:2222->22/tcp, [::]:2222->22/tcp   suse-web-test
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker exec suse-web-tes
t pgrep -a sshd
1 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker login --username
meghanaaa08

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:
Login Succeeded
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> sudo docker push meghanaaa08/
suse-web-lab:v1
The push refers to repository [docker.io/meghanaaa08/suse-web-lab]
0e4a321f46ec: Pushed
6a32c679a29a: Pushed
cf1a5487baaf: Pushed
v1: digest: sha256:94cd5d23eeeba75690d8e06781cef11524a5b5c75252936924b479b611622
c75 size: 855
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo>
```



Create the Network and The Containers Ping the Containers From Another Using the Names:

```
b-network alpine:latest sh
77a323ce0c1b56720de298e3e8decad9cb681cca29da36cc920a729960fe512d
ec2-user@ip-172-31-41-183:~> sudo docker run -dit --name container2 --network la
b-network alpine:latest sh
783834e0bcba2c7b1ac64d63c30012b28f777f2db8fa0109ec8936c9edf13de8
ec2-user@ip-172-31-41-183:~> sudo docker ps
CONTAINER ID   IMAGE                          COMMAND              CREATED
    STATUS          PORTS                                  NAMES
783834e0bcba   alpine:latest                  "sh"                 8 seconds ago
    Up 7 seconds                                           container2
77a323ce0c1b   alpine:latest                  "sh"                 15 seconds ag
o   Up 14 seconds                                          container1
cf00cd2a781e   meghanaaa08/suse-web-lab:v1    "/usr/sbin/sshd -D"   6 minutes ago
    Up 6 minutes    0.0.0.0:2222->22/tcp, [::]:2222->22/tcp   suse-web-test
ec2-user@ip-172-31-41-183:~> sudo docker exec container1 ping -c 4 container2
PING container2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.266 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.083 ms
64 bytes from 172.18.0.3: seq=2 ttl=64 time=0.076 ms
64 bytes from 172.18.0.3: seq=3 ttl=64 time=0.089 ms

--- container2 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.076/0.128/0.266 ms
ec2-user@ip-172-31-41-183:~> sudo docker exec container2 ping -c 4 container1
PING container1 (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: seq=0 ttl=64 time=0.079 ms
64 bytes from 172.18.0.2: seq=1 ttl=64 time=0.140 ms
64 bytes from 172.18.0.2: seq=2 ttl=64 time=0.131 ms
64 bytes from 172.18.0.2: seq=3 ttl=64 time=0.080 ms

--- container1 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.079/0.107/0.140 ms
ec2-user@ip-172-31-41-183:~>
```

32.236.201.54 - PuTTY

```
ec2-user@ip-172-31-41-183:~/docker-image-lab-demo> cd
ec2-user@ip-172-31-41-183:~> whoami
ec2-user
ec2-user@ip-172-31-41-183:~> cat etc/os-realese
cat: etc/os-realese: No such file or directory
ec2-user@ip-172-31-41-183:~> cat etc/os-release
cat: etc/os-release: No such file or directory
ec2-user@ip-172-31-41-183:~> cat etc/os_release
cat: etc/os_release: No such file or directory
ec2-user@ip-172-31-41-183:~> sudo docker network create lab-network
2447859ad2434a030d0f1eb5e14e29117b6e5eb97202b8369c19619e01dc0f41
ec2-user@ip-172-31-41-183:~> sudo docke5r network ls
sudo: docke5r: command not found
ec2-user@ip-172-31-41-183:~> sudo docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
204b9cb5d9db   bridge        bridge    local
94abd501d890   host          host      local
2447859ad243   lab-network   bridge    local
170dfe3033c8   none          null      local
ec2-user@ip-172-31-41-183:~> sudo docker run -dit --name container1 --network la
b-network alpine:latest sh
77a323ce0c1b56720de298e3e8decad9cb681cca29da36cc920a729960fe512d
ec2-user@ip-172-31-41-183:~> sudo docker run -dit --name container2 --network la
b-network alpine:latest sh
783834e0bcba2c7b1ac64d63c30012b28f777f2db8fa0109ec8936c9edf13de8
ec2-user@ip-172-31-41-183:~> sudo docker ps
CONTAINER ID   IMAGE                          COMMAND               CREATED
    STATUS          PORTS                                  NAMES
783834e0bcba   alpine:latest                  "sh"                  8 seconds ago
    Up 7 seconds                                           container2
77a323ce0c1b   alpine:latest                  "sh"                  15 seconds ag
o   Up 14 seconds                                          container1
cf00cd2a781e   meghanaaa08/suse-web-lab:v1   "/usr/sbin/sshd -D"   6 minutes ago
    Up 6 minutes    0.0.0.0:2222->22/tcp, [::]:2222->22/tcp   suse-web-test
ec2-user@ip-172-31-41-183:~> sudo docker exec container1 ping -c 4 container2
PING container2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.266 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.083 ms
64 bytes from 172.18.0.3: seq=2 ttl=64 time=0.076 ms
64 bytes from 172.18.0.3: seq=3 ttl=64 time=0.089 ms

--- container2 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.076/0.128/0.266 ms
ec2-user@ip-172-31-41-183:~> sudo docker exec container2 ping -c 4 container1
PING container1 (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: seq=0 ttl=64 time=0.079 ms
64 bytes from 172.18.0.2: seq=1 ttl=64 time=0.140 ms
64 bytes from 172.18.0.2: seq=2 ttl=64 time=0.131 ms
```